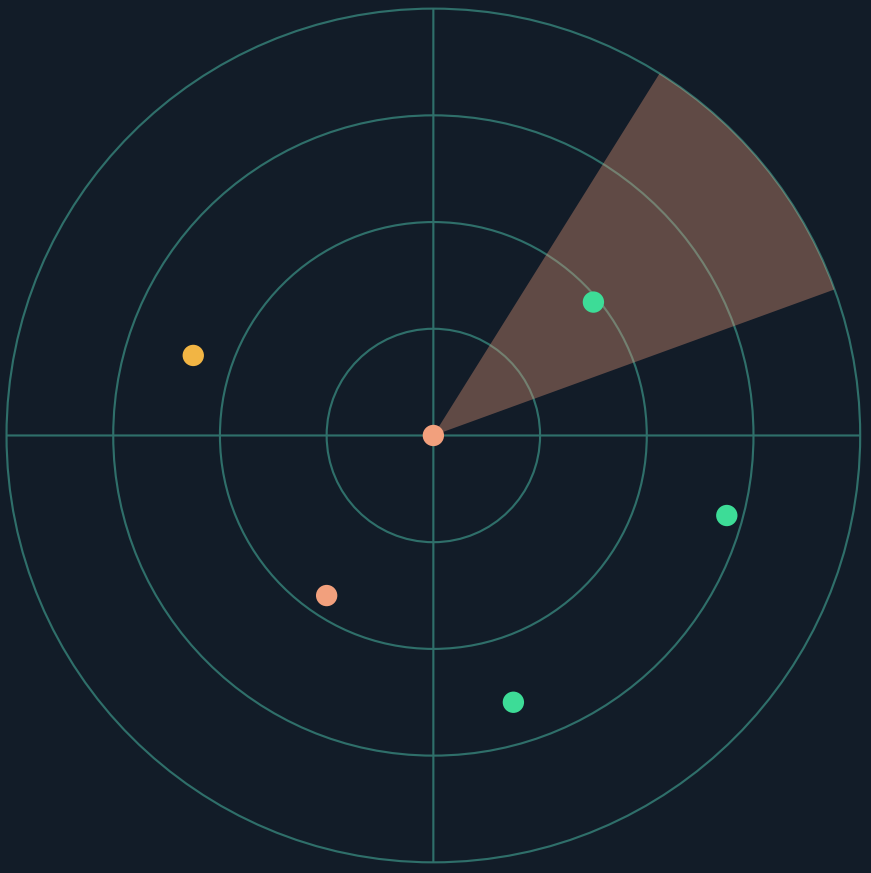

Crosshair

Find the flaws. Win the client.

Pitch & tech prep guide
The pitch, the demo, how it works, the full tech stack,
the architecture, real numbers, and answers to judge questions.

github.com/srutesh18/CrossHair - Next.js 15 - Supabase - Crawlee - three.js

01 - THE IDEA

Crosshair in one sentence

Crosshair finds businesses on its own, audits their websites for the exact problems you fix, spots who just raised money or hired a new leader, and writes the email with proof in every line.

Who it's for

Web design agencies, SEO consultants, AI-automation builders, freelance developers, fractional CMOs: anyone who sells a service to businesses and today finds clients by cold, generic outreach.

The problem

- Finding clients is manual: scroll forums, google companies, open their sites one by one.
- Cold emails are generic ("I can help with your website"), so they get ignored.
- Timing is luck: you email a company the week they cut budget, not the week they raised.

Our answer: Pain + Trigger + Reach

Signal	What Crosshair finds	Why it matters
Pain	Real flaws on their own website that your services fix, with evidence ("homepage took 7.4s to load").	The email is about *their* problem, provable in one click.
Trigger	Fresh news: funding, a new executive, a hiring push, an expansion.	They have budget and a new decision-maker *now*.
Reach	Public email, phone, contact form or social profile from their own site.	You can actually contact them, legitimately.

What makes it different

- **Fully automatic discovery.** No lead lists to buy. One click scans Reddit, Hacker News, community forums and startup news.
- **Evidence, not adjectives.** Every flaw quotes what was measured on the site.
- **Timing built in.** News triggers raise the score of companies that just raised money or hired a leader.
- **Honest by design.** Public data only, robots.txt respected, nothing sent without a human, an opt-out in every email.

02 - SAY IT

The pitch scripts

Learn the 60-second version word for word; use the 3-minute version on stage.

60-second pitch

> *"Every agency and freelancer has the same problem: finding clients means hours of scrolling and generic cold emails nobody reads. Crosshair fixes that. Tell it what you sell, say web design and SEO. Press Scan. It finds business owners sharing their websites on Reddit and Hacker News, and companies in the news that just raised money or hired a new leader. It audits every website: is it mobile-friendly, how fast does it load, is Google missing its description, can customers book online. It finds the public email on their site, scores each company on pain, trigger and reach, and writes the email with the evidence in it: 'Congrats on the Series A. Your homepage takes 7 seconds to load on mobile.' Then it tracks your Day 3 and Day 7 follow-ups. Crosshair: find the flaws, win the client."*

3-minute pitch (structure)

Time	Beat	What to say / show
0:00	Hook	"How many of you have sent a cold email that started with 'I hope this finds you well'? It didn't."
0:15	Problem	Finding clients is manual, emails are generic, timing is luck. Agencies spend more time prospecting than delivering.
0:40	Solution	Crosshair: automatic discovery + website audit + news triggers + written outreach. Pain, trigger, reach.
1:00	Live demo	Radar: Scan now, 3D radar fills, Top opportunities with 'Why now'. Open a company: flaws with evidence. Add to pipeline, open the composer, show the written email, Open in Gmail.
2:20	How it's built	Next.js + Supabase with row-level security, Crawlee crawler, Google News & startup feeds, three.js. Public data only, robots.txt respected.
2:40	Proof	"In our test, one scan found 86 companies in 86 seconds, 26 of them with fresh funding or leadership news."
2:50	Close	"Crosshair. Find the flaws. Win the client." Pause. Thank you.

Delivery tips. Lead with the email example; it is the most concrete moment. Say numbers we measured, never rounded up. If the demo stalls on a slow scan, narrate what it is doing ("it's verifying each company's website right now") and move to the pre-scanned results.

03 - SHOW IT

Live demo script

Rehearse this exact path. Run one scan before you go on stage so results are already there.

#	Where	What to do
1	Landing page (/)	Show the 3D globe and the headline. Click **Open dashboard**.
2	Radar (/finder)	Point at the sidebar: live mini-radar, Today card, weekly goal. Press **Scan now**. The 3D radar beam speeds up; each source reports as it finishes.
3	Top opportunities	Read one card aloud: score, **Why now** (funding headline), two flaws with evidence, the email found.
4	Trigger feed	"These companies raised money or hired a leader this month."
5	Table	Filter **Just raised money**, then **Has email**. Click a row to expand every flaw with its evidence.
6	Add to pipeline	Tick 2-3 rows, press **Add to pipeline & email**.
7	Pipeline	Show the 3D funnel and the board. Click a card: the composer has written Day 0 from their real flaws and news.
8	Send	Press **Open in Gmail** (a real draft opens), then **Yes, I sent it**. The Day 3 follow-up is now scheduled.
9	Templates	Show the live preview updating as you type. Mention the opt-out line.
10	Cmd/Ctrl + K	Open the command palette to jump around. End on the radar.

Backup plan

- **Before going on stage:** sign in, run one full scan, add 3 companies to the pipeline, send one Day 0 via Gmail so the board has every column filled.
- **Bad Wi-Fi:** the Radar, Top opportunities, table and pipeline all load from the database; only *new* scans need the internet.
- **Reddit rate-limited:** say so. The scan still finishes with Hacker News, communities and news; the source line shows the reason.
- **A scan takes ~1.5 min** because news websites are verified one by one: start it, talk through the audit, come back.

04 - UNDERSTAND IT

How Crosshair works

Five stages, each with a clear input and output.

Stage	Input -> output	Where in code
1. Discover	Public posts and news -> company websites. Reddit business/feedback communities, Hacker News 'Show HN' launches, Bubble/n8n/Make forums, and startup news feeds.	lib/discovery.ts, lib/sources/*
2. Trigger	News headline -> company + trigger kind (Funding / New executive / Hiring / Expansion). Website found by verified domain guessing.	lib/sources/companyNews.ts
3. Audit	Website -> flaws with evidence + public contacts. Homepage timed; Crawlee visits contact/about pages; robots.txt honoured.	lib/audit/audit.ts, checks.ts, contacts.ts
4. Score	Flaws + trigger + contact -> 0-100 opportunity score, matched to the services you sell.	lib/audit/score.ts
5. Outreach	Company -> Day 0 / 3 / 7 emails from templates; pipeline stages; follow-up scheduling.	lib/outreach.ts, lib/templates.ts

The scoring formula

Part	Points	Rule
Pain	up to 75	For each flaw in an area your services cover: high 25, medium 12, low 5 (capped at 75).
Trigger	up to 15	Funding or new executive within 30 days: +15; hiring/expansion: +10; halved after 30 days, none after 90.
Reach	10	+10 when a public email or phone was found on their site.

What the audit checks

Service	Flaws detected
Web design	No website - site down - no HTTPS - not mobile-friendly (no viewport tag) - slow homepage (timed) - poor PageSpeed (optional) - copyright year 2+ years old - broken links (404) - outdated WordPress / jQuery / Flash - hard to contact
SEO	No / weak page title - no meta description - no H1 - noindex (hidden from Google) - no sitemap - images without alt text - no schema.org data - very little text - no analytics
AI automation	No online booking (only for appointment businesses, judged from title, description, headings, menu) - no chat, chatbot or WhatsApp
App development	No customer login, online ordering or mobile app (only for shops, restaurants, member businesses)

Accuracy safeguards worth mentioning: English-only filter; platform links skipped (Reddit, GitHub, app stores); sellers and job ads dropped; a news company's website is only accepted if the site's own title names every word of the company, and one-word names must also match the headline's topic (so 'TODAY raises 2.8M' never resolves to today.com). Investment funds are skipped.

05 - KNOW THE STACK

Tech stack, and why each piece

Be ready to say what each technology does in Crosshair and why we picked it.

Layer	Technology	What it does here	Why we chose it
Framework	Next.js 15 (App Router), React 19, TypeScript (strict)	Pages, server components, and API route handlers in one app.	One codebase for UI + API; server components keep secrets on the server.
Styling	Tailwind CSS v4 + custom design tokens	Layout, colours, cards, animations.	Fast iteration; consistent design system.
Database	Supabase Postgres	Companies, pipeline, emails, scans, workspaces.	Managed Postgres with auth and APIs built in.
Auth	Supabase Auth (email + password, PKCE)	Sign up, sign in, email confirmation.	Secure sessions with no auth server to run.
Security	Row Level Security (RLS) policies	Every table limited to the signed-in user's workspace.	Data isolation enforced in the database itself.
Validation	Zod	Every API request body is validated.	Bad input is rejected before it touches data.
Crawling	Crawlee (CheerioCrawler) + cheerio	Visits contact/about pages; parses HTML for flaws and contacts.	Open-source (Apify), honours robots.txt, fast HTML parsing.
Sources	Reddit RSS/API, Hacker News Algolia API, Discourse search, RSS feeds (TechCrunch, tech.eu, BetaKit, Inc42, PR Newswire), Google News RSS	Discovery and news triggers.	Free, public, no scraping behind logins.
Entity resolution	Verified domain guessing + Wikidata API	News company name -> correct website.	No paid enrichment API; verification prevents wrong matches.
Classifier	Typesafe AI (jev-latest)	Calibrated second opinion on categories/ratings.	Returns probabilities; we only trust it when confident.
Generation	Groq (openai/gpt-oss-120b), optional	Original AI scoring/drafting path.	Fast inference; current email flow works without it (templates).
RAG	pgvector + Transformers.js (all-MiniLM-L6-v2)	Product docs embedded locally for grounded drafting.	Free, local embeddings; vector search inside Postgres (HNSW).
3D / UI	three.js + React Three Fiber + drei	3D radar, globe, pipeline funnel.	Interactive WebGL inside React components.
Email	Nodemailer (SMTP) + Gmail/Outlook/mailto drafts	Send now, or open a ready draft in the user's email.	Works with zero setup; SMTP when configured.
Exports	papaparse, exceljs, @react-pdf/renderer	CSV / XLSX / PDF exports.	Reliable open-source exporters.
Dev tools	Supabase MCP, migrations runner, headless Chrome	Applied migrations, checked security advisors, screenshot QA.	Repeatable setup and visual verification.

06 - ARCHITECTURE

How a scan travels through the system

Step	What happens
1	Browser: **Scan now** calls `POST /api/discover` once per source (Reddit, HN, communities, news) so each reports separately.
2	Server: `requireSession()` verifies the user; every database call runs as that user, so RLS scopes rows to their workspace.
3	`discoverCompanies()` fetches public posts/news, extracts the owner's website (or resolves it from the headline), ranks businesses first, de-duplicates.
4	New companies are inserted (status *new*). The browser loops `POST /api/companies/audit` in batches of 6 (3 in parallel) with a progress bar.
5	`auditWebsite()`: robots.txt check, timed homepage fetch, Crawlee on contact pages, broken-link and sitemap checks, flaw rules, contacts.
6	`scoreCompany()` adds pain + trigger + reach; the row becomes *audited*; the radar, panels and table refresh.
7	**Add to pipeline** creates an *outreach* row; the composer renders templates server-side; sending logs to *outreach_emails* and schedules the next step.

Key files

Path	Purpose
app/(app)/finder/page.tsx	Radar dashboard: hub, panels, filters, table
app/(app)/pipeline/page.tsx	Pipeline board + composer
app/(app)/templates/page.tsx	Email sequence editor with live preview
app/api/discover, companies/*, pipeline/*, templates	API route handlers (Zod-validated)
lib/discovery.ts	Automatic discovery across sources
lib/sources/companyNews.ts	News triggers + website resolution
lib/audit/*	Auditor, flaw checks, contact extraction, scoring
lib/outreach.ts, lib/templates.ts	Pipeline logic and email templates
components/clients/Radar3D.tsx, components/three/*	3D radar, globe, funnel
supabase/migrations/*.sql	Versioned schema (11 migrations, all applied)

Database tables

Table	Holds
orgs / org_members	Workspaces and who belongs to them (each signup gets its own)
companies	Website, contacts, flaws (with evidence), score, trigger, where found, status
outreach / outreach_emails	Pipeline stage, next step and date, notes / every email sent

Table	Holds
crawl_runs	History of scans with per-source stats
docs / doc_chunks	Product docs and their vector embeddings (RAG)
leads, lead_scores, email_drafts, signals	Earlier complaint-based flow (kept, not in the main navigation)

07 - TRUST

Security, privacy and ethics

- **Public data only.** Posts people chose to publish, news articles, and companies' own websites. Nothing behind a login.
- **robots.txt respected.** Sites that disallow crawlers are skipped and marked as such.
- **Polite crawling.** Requests spaced out, small batches, one retry on rate limits, descriptive User-Agent.
- **Workspace isolation.** Row Level Security on every table; even our own API cannot read another workspace's rows.
- **Secrets stay on the server.** Keys live in .env.local (git-ignored); the repo was scanned for keys before every push.
- **Human in the loop.** Nothing is emailed automatically; the user reviews and sends.
- **Opt-out.** Every default email ends with "reply 'no' and I won't follow up"; replied/won/lost stops follow-ups.
- **Database hardening.** Supabase security advisor checked: security-definer functions locked down, policies scoped to signed-in users.

08 - PROOF

Real numbers from our test runs

Use these exactly; they were measured during development.

111 company websites found in 12 s (Reddit, HN, forums)	**86 / 86 s** companies from all 4 sources in one scan	**26** of them with fresh news triggers
128 funding / leadership headlines parsed (30 days)	**16** news websites verified, all correct on review	**8 in 3 s** websites audited in parallel

Examples seen in real runs: "Cinch Home Services appoints a new Chief Marketing Officer" (cinchhomeservices.com); "Healthleap raises $38M" (healthleap.ai); an audit of sniptura.com scoring 66 (no meta description, no main heading, hard to contact).

09 - BE READY

Likely judge questions, and good answers

Question	Answer
Isn't this just spam with extra steps?	The opposite. We email fewer companies, only ones with a real, provable problem you fix, using an address they published, with an opt-out and a human pressing send.
Is scraping legal?	We only read public pages, honour robots.txt, never log in, and use official APIs/RSS where they exist (Hacker News Algolia, Reddit RSS/API, Discourse search, news RSS).
How do you find the company's email?	From their own website: mailto links, visible text, schema.org data, even Cloudflare-protected addresses. If none, we show their contact form or the post they wrote.
How do you know the news is about the right company?	Verified matching: a candidate domain is accepted only if the site's own title names every word of the company, plus a topic check for one-word names, with Wikidata as a checked fallback.
Why not just use an LLM for everything?	Speed, cost and trust. Rules give evidence we can quote and explain. AI (Typesafe, Groq) is used where it adds value, and the core flow works even if a provider is down.
What's your accuracy?	We measured and fixed failure cases: wrong-company news matches, sellers posing as buyers, job ads, script-embedded junk emails. Be honest: indie sites from Reddit/HN are startups more often than local businesses.
How does it scale?	Audits run in small batches within serverless time limits; Postgres with indexes and RLS; adding a Google Places key adds local-business search; Reddit API credentials raise rate limits ~100x.
Business model?	Per-seat SaaS for agencies and freelancers, with usage tiers by companies scanned per month; higher tiers add automated sending, calls (Sarvam) and LinkedIn (n8n).
What was hardest?	Data quality: turning messy public posts and headlines into the right company with the right website. Each safeguard came from a real failure we saw in testing.
What's next?	Email, call (Sarvam AI) and LinkedIn (n8n) automation, scheduled daily scans, Google Places for local businesses, AI-written personalised emails, deployment.

Know our honest limits (judges respect this): Reddit allows ~1 unauthenticated search per minute (fix: free Reddit app credentials); Google Places search needs a Google API key; direct sending needs SMTP details (Gmail/Outlook drafts work without); the Groq key we had was rejected, so emails are template-based today.

10 - WRAP UP

Roadmap, glossary and run checklist

Roadmap

Now (built)	Next	Later
Automatic discovery (4 sources) - website audit - news triggers - scoring - 3D radar - pipeline with Day 0/3/7 - templates - Gmail/Outlook drafts - command palette	SMTP sending on by default - scheduled daily scans - Google Places local search - Reddit API credentials - AI-personalised emails	Sarvam AI voice calls - n8n LinkedIn DMs - reply detection from the inbox - team workspaces - public deployment

Glossary

Term	Meaning
RLS (Row Level Security)	Postgres rules that decide which rows a user may read or write; enforced by the database.
Server component	React component rendered on the server; can read secrets and query the database safely.
Route handler	A Next.js API endpoint (app/api/.../route.ts).
RAG	Retrieval-augmented generation: fetch relevant documents first, then let the AI write from them.
pgvector / HNSW	Postgres extension for vector similarity search / the fast index it uses.
Embedding	A list of numbers representing meaning; similar texts have similar embeddings.
Crawler	Program that visits web pages automatically (we use Crawlee).
robots.txt	A site's file telling crawlers what they may not visit.
RSS	A public feed format news sites publish for readers and apps.
Trigger	A recent event (funding, new executive) that makes now the right time to reach out.
WebGL / three.js	Browser 3D graphics / the library we use to draw the radar, globe and funnel.
SMTP	The protocol for sending email from a mailbox.

Run checklist

Step	Command / action
Start the app	cd crosshair-next && npm run dev -> http://localhost:3000
Check configuration	http://localhost:3000/api/health (ok: true)
Database	Already migrated; for a new project: npm run db:migrate
Optional keys (.env.local)	GOOGLE_API_KEY - REDDIT_CLIENT_ID / SECRET - SEND_ENABLED + SMTP_* + SEND_FROM + UNSUBSCRIBE_URL
Before the pitch	Sign in, run one scan, add 3 companies to the pipeline, send one Day 0 via Gmail